

THE INTERN DIARIES

Intern Report

SQL Window Functions Intro

WEEK 3 | DAILY JOURNAL ENTRY



LEARNING BY DOING, ONE DAY AT A TIME

Online Retail II (RETAIL COMMERCE)

1. How is the business performing in terms of revenue, orders, customers and order value?

Answer: Management needs a high-level view of the company's overall sales performance. Before analyzing individual countries, products, or customers, basic business KPIs must be established.

What it does:

The query calculates five headline KPIs:

- **Total number of orders**
- **Number of sales transaction lines**
- **Total revenue**
- **Average Order Value (AOV)**
- **Number of unique customers**

Revenue is calculated as:

Quantity × Price

Average Order Value is calculated as:

Total Revenue ÷ Number of Orders

2. Which countries generate the highest revenue and how many orders do those markets contribute?

Answer: The company operates across multiple countries, but revenue may not be distributed equally across geographic markets. Identifying the strongest markets helps management understand where sales are concentrated.

What it does

The query:

- **Groups sales by country**
- **Calculates revenue for each country**
- **Counts the number of orders from each country**
- **Sorts countries by revenue**
- **Returns the top 10 countries**

3. How does revenue change from month to month, and when do significant increases or decreases occur?

Answer: Looking only at total revenue does not show how business performance changes over time. Management needs to identify growth, decline, seasonal patterns, and unusually strong or weak months.

What it does

The query:

- 1. Groups transactions by month.**
- 2. Calculates monthly revenue.**
- 3. Uses the LAG() window function to retrieve the previous month's revenue.**
- 4. Calculates the month-over-month percentage change.**

The formula is:

$$\text{MoM \%} = \frac{(\text{Current Month Revenue} - \text{Previous Month Revenue})}{\text{Previous Month Revenue}} \times 100$$

4. Which products generate the most revenue, and how many units of those products are being sold?

Answer: A company may sell thousands of products, but a relatively small number of products can contribute a large proportion of total revenue. Identifying these products helps management understand product-level performance.

What it does

The query:

- **Groups transactions by StockCode**
- **Retrieves a representative product description**

- **Calculates total revenue for each product**
- **Calculates total units sold**
- **Sorts products by revenue**
- **Returns the top 10 products**

5. How concentrated is the company's revenue among its highest-value customers?

Answer: Revenue concentration is important because a business may depend heavily on a relatively small group of high-value customers.

If a small percentage of customers generates a large percentage of revenue, losing those customers could have a significant business impact.

What it does

The query:

- 1. Calculates total revenue for each customer.**
- 2. Ranks customers into 100 groups using NTILE(100).**
- 3. Calculates the percentage of total revenue generated by:**
 - The top 1% of customers
 - The top 10% of customers

6. How much sales value is being lost through returns, and what proportion of sales does that represent?

Answer: Revenue alone does not provide a complete picture of business performance. Returns and cancellations can reduce the effective value of sales and may indicate issues with products, fulfillment, or customer expectations.

What it does

The query calculates:

- **Total sales value**
- **Total returned value**
- **Return value as a percentage of sales**

The return rate is calculated as:

Returned Value ÷ Sales Value × 100

7. What does a typical order look like, and how large can high-value orders become?

Answer: Average order value alone may hide significant differences between small and very large orders. Businesses therefore need multiple measures to understand typical and unusually large purchases.

What it does

First, the query calculates the value and number of units for each order.

It then calculates:

- **Average units per order**
- **Average order value**
- **Median order value**
- **90th-percentile order value**
- **Maximum order value**

The median is particularly useful because it represents the middle order and is less affected by extreme outliers.

8. What proportion of customers return to make additional purchases?

Answer: Acquiring new customers can be more difficult than retaining existing customers. Therefore, understanding how many customers make multiple purchases is an important measure of customer loyalty and engagement.

What it does

The query:

1. **Groups transactions by customer.**
2. **Counts the number of unique orders for each customer.**
3. **Identifies customers with more than one order.**
4. **Calculates the percentage of customers who are repeat purchasers.**
5. **Calculates average orders per customer.**

9. Which customers have generated the highest lifetime revenue for the business?

Answer: Not all customers contribute equally to revenue. Identifying the highest-value customers allows the business to understand its most financially significant customer accounts.

What it does

The query calculates, for every customer:

- **Total lifetime revenue**
- **Number of orders**

It then sorts customers by lifetime revenue and returns the top 20.

10. Which days of the week generate the most revenue and orders?

Answer: Customer purchasing behavior may vary depending on the day of the week. Understanding these patterns can help the business allocate resources and plan marketing activities.

What it does

The query:

- **Extracts the weekday from each transaction date.**
- **Groups transactions by weekday.**
- **Calculates revenue for each weekday.**
- **Counts the number of orders.**
- **Sorts the weekdays according to revenue.**

11. Which products generate the largest financial value of returns?

Answer: Identifying products with high return activity can reveal potential product-level problems. However, the number of returned units alone may not indicate financial impact, so the query also measures return value.

What it does

The query:

- **Filters negative-quantity transactions.**
- **Groups returns by stock code.**
- **Calculates units returned.**

- **Calculates the monetary value of returns.**
- **Identifies the top 10 products based on return value.**

12. How can customers be segmented according to how recently, how frequently, and how much they purchase?

Answer: A simple customer ranking does not fully describe customer behavior. Businesses need to distinguish between customers who purchase recently, purchase frequently, and generate significant revenue.

RFM analysis provides a structured way to segment customers based on three dimensions:

R = Recency , F = Frequency, M = Monetary

What it does

The query calculates:

Recency

Number of days since the customer's most recent purchase

Frequency

Number of distinct orders made by the customer

Monetary

Total revenue generated by the customer

Each measure is then divided into five NTILE() groups.

The resulting scores are:

R5 → Recency score

F5 → Frequency score

M5 → Monetary score

The query then groups customers by their RFM scores and shows the number of customers and average monetary value in each combination.